Seasonal Forcing Analysis for Bd Model Parameters in *Litoria aurea*

Overview

This report assesses how each parameter in the *L. aurea* Bd compartmental ODE model is affected by seasonal temperature and behavioral cycles, drawing on species-specific experimental data and the broader Bd literature already compiled. All analysis is framed around the Sydney, NSW climate where the largest remaining *L. aurea* populations occur: winter (June–August) mean temperatures ~8–12°C, summer (December–February) ~24–30°C, with the Southern Hemisphere seasons offset by six months from the Northern Hemisphere convention.

The original ODE includes a commented-out seasonal forcing term of the form $\boldsymbol{\alpha_1 \cdot (1 + \sigma \cdot \cos(2\pi t))}$, where σ is the amplitude of seasonal variation. This report evaluates which parameters should receive such forcing, at what amplitude, and with what phase.

β (Disease-Induced Mortality) — THE Most Seasonally Sensitive Parameter

Seasonal range: ~10-fold (0.002–0.025 day^{-1})

This is where the evidence is strongest and most dramatic. Campbell et al. (2019, Sci. Rep. 9:83) measured *L. aurea* survival directly at two temperatures that roughly correspond to Sydney's seasonal extremes:

- **At 12°C (winter analog): 30% survival** after 48 days → $\beta \approx 0.025$ day^{-1}
- **At 24°C (summer analog): 90% survival** after 48 days → $\beta \approx 0.002$ day^{-1}
- **Uninfected controls: 95% survival** at both temperatures

This translates to more than a **12-fold seasonal swing** in disease-induced mortality. The mechanism is a compounding effect operating on both sides of the host–pathogen interaction. At cold temperatures, frog immune function is suppressed — reduced lymphocyte activity, slower antimicrobial skin peptide secretion — as demonstrated in anurans generally by Maniero & Carey (1997, J. Comp. Physiol. B 167:256–263). Simultaneously, Campbell et al. found that Bd infection loads in *L. aurea* were significantly higher and increased faster at 12°C than at 24°C, which is counterintuitive given Bd's in-culture growth optimum of 17–25°C (Piotrowski et al. 2004, Mycologia 96:9–15). The explanation is that the collapse of host defenses in the cold overwhelms the pathogen's own somewhat slower replication at suboptimal temperatures, resulting in net higher loads at cooler temperatures. Waddle's field observations reinforce this starkly: he described populations in Sydney as experiencing mass mortality every winter, with dead and dying frogs widespread in cooler months.

Recommended seasonal forcing: Peak β in mid-winter (July). A cosine term with amplitude **σ_β ≈ 0.8–0.9** would capture the ~10× range. If using the form $\beta(t) = \beta_0 \cdot (1 + \sigma_\beta \cdot \cos(2\pi t/365))$, set β_0 to approximately 0.010–0.013 day^{-1} (the annual geometric mean) and σ_β ≈ 0.85, so that β oscillates between ~0.002 (summer trough) and ~0.024 (winter peak).

If you only add seasonal forcing to one parameter, β is the one to choose.

α (Transmission Rate) — Strongly Seasonal, but with Competing Mechanisms

Seasonal range: ~3–5×, with complex phasing

Transmission is seasonal for *L. aurea* through at least three interacting pathways, and these pathways do not all peak in the same season.

Pathway 1 — Zoospore ecology favors winter transmission. Piotrowski et al. (2004) showed that while Bd zoosporangia develop faster at 17–25°C, zoospores produced at 7–10°C remain motile and infectious for longer periods. The net effect is that winter zoospores are slower to produce but more persistent in the environment once released, increasing per-contact transmission probability. Johnson & Speare (2003, Emerg. Infect. Dis. 9:922–925) demonstrated Bd survival in water for 3–7 weeks at 23°C; survival would be extended further at cooler winter water temperatures.

Pathway 2 — Breeding aggregation favors spring/summer transmission. *L. aurea* breeding aggregations occur in spring and summer (approximately October–February in NSW), concentrating frogs in shared aquatic habitat and dramatically increasing contact rates. Rachowicz & Briggs (2007, J. Anim. Ecol. 76:711–721) validated density-dependent transmission for Bd, meaning that α effectively increases during breeding season due to higher local densities, even as temperature-driven Bd viability may be decreasing.

Pathway 3 — Bd prevalence peaks in cooler months. Stockwell et al. (2016, FEMS Microbiol. Lett. 363:fnw111) and the broader *L. aurea* field literature show that Bd prevalence and infection intensity in wild populations peak in the cooler months (autumn/winter) and drop in summer. This reflects the net outcome of the two pathways above: even though contact rates are lower in winter, per-contact transmission probability is high enough to drive higher prevalence.

Net seasonal pattern: The product of per-contact transmission probability (peaking late autumn/winter) and contact rate (peaking spring/summer) likely gives realized transmission a broad peak from late autumn through spring, with a trough in mid-to-late summer when both Bd viability is reduced by heat and frogs are dispersed post-breeding. A simple cosine approximation peaking in late autumn/early winter (May–June) is a reasonable first-order approximation.

Recommended seasonal forcing: Amplitude **σ_α ≈ 0.4–0.6**, with phase set so that peak α occurs approximately 1–2 months before peak β (reflecting the lag between peak transmission and peak

mortality as infection loads build). If β peaks in July, α should peak around May–June.

ζ (Frog Sauna Entry/Clearance Rate) — Seasonal by Design and by Physics

Seasonal range: ~50–100% of maximum, with critical winter reliability concern

This parameter is inherently seasonal because the frog sauna's thermal mechanism depends entirely on solar irradiance. Waddle & Shine's construction guide explicitly states that shelters "have the best impact in fall/autumn and winter when chytrid thrives" — but this is paradoxically when solar angles are lowest and day lengths shortest in Sydney (~10 hours of daylight in June vs. ~14.5 hours in December).

The critical question is whether the shelters reach ⩾28–30°C in winter. Sydney winter daytime air temperatures average ~17°C, and clear-sky solar radiation on a well-positioned greenhouse can add 10–15°C internally, pushing temperatures to ~27–32°C on sunny winter days. However, on overcast or rainy days — which are common in Sydney winters — the sauna may not reach therapeutic temperatures at all. So ζ is effectively **near zero on overcast winter days** and maximal on sunny summer days (when it is least needed, since Bd is already suppressed by ambient warmth).

Waddle et al. (2024, Nature 631:344–349) demonstrated that even with this constraint, unshaded shelters in the mesocosm experiment produced roughly **100-fold lower Bd infection intensity** compared to shaded controls over cooler months. This suggests that the sunny-day fraction in Sydney winters is sufficient for meaningful therapeutic benefit — but ζ should realistically be modeled as lower in winter than summer, rather than constant.

Recommended seasonal forcing: Amplitude **σ_ζ ≈ 0.3–0.5**, with peak in summer (December–January) and trough in winter (June–July). A central ζ_0 of ~0.12 day^{-1} with σ_ζ = 0.4 would give summer values of ~0.17 day^{-1} and winter values of ~0.07 day^{-1}. This captures the reduced but still nonzero effectiveness of saunas during the critical winter period.

δ (Cadaver Bd Decay Rate) — Moderately Seasonal

Seasonal range: ~2–3× (0.020–0.050 day^{-1})

Johnson & Speare's (2003, 2005) data on Bd environmental persistence were collected in Australia and show clear temperature and moisture sensitivity. In water at 23°C, Bd survived 3–7 weeks; survival extends considerably at cooler winter water temperatures. Separately, Bd cannot survive desiccation (Longcore et al. 1999), and Sydney summers have higher evaporation rates and more frequent drying of ephemeral wetland margins, meaning cadavers and wet substrates dry out faster, killing Bd rapidly.

In winter, cooler and moister conditions extend Bd viability on cadavers. The seasonal range for δ spans approximately:

- **Summer:** $\delta \approx 0.050$ day^{-1} (rapid desiccation and thermal die-off; ~20-day cadaver infectiousness)
- **Winter:** $\delta \approx 0.020$ day^{-1} (moist, cool conditions; viability extending to ~50 days)

This is a meaningful but not dramatic swing compared to β.

Recommended seasonal forcing: Amplitude **σ_δ ≈ 0.3–0.4**, with peak δ (fastest decay, shortest cadaver infectivity) in summer and trough in winter. Phase is opposite to β: δ is highest when temperatures are highest.

γ (Cadaver-to-Susceptible Transmission) — Moderately Seasonal, Tracks δ Inversely

Seasonal range: ~2–3×, inversely correlated with temperature

This parameter scales with both the duration of cadaver infectivity (proportional to 1/δ) and the zoospore shedding rate from cadavers, both of which are temperature-dependent. In winter, cadavers remain infectious longer and shed zoospores into cooler water where those zoospores persist longer. In summer, cadavers dry out and Bd dies rapidly. γ is approximately proportional to 1/δ, making it inversely seasonal with temperature — higher in winter, lower in summer — with a similar 2–3× range.

Recommended seasonal forcing: Mirror δ's seasonal pattern but inverted. Amplitude **σ_γ ≈ 0.3–0.4**, peaking in winter when cadaver transmission is most efficient.

ξ (Treatment Capture Rate) — Strongly Seasonal Due to Frog Behavior

Seasonal range: effectively ~5–8×, approaching zero in winter

L. aurea activity is highly seasonal. During the breeding season (spring/summer, October–February), frogs aggregate at wetlands, call loudly, and are conspicuous, making capture relatively efficient. Recapture probabilities during the active season are typically 0.10–0.30 per weekly survey (Beranek et al. 2021, J. Environ. Manage. 291:112742). In winter, *L. aurea* becomes cryptic, retreating to terrestrial refugia under rocks, logs, and debris, and detection probability drops substantially. Many *L. aurea* field studies only survey during the active season because winter captures are so difficult.

This means ξ is effectively **near zero in winter and maximal in spring/summer**:

- **Winter (June–August):** $\xi \approx 0.003–0.005$ day^{-1} (frogs nearly undetectable)
- **Summer (November–February):** $\xi \approx 0.03–0.04$ day^{-1} (frogs aggregated, vocal, conspicuous)

This creates a frustrating management paradox: frogs are easiest to capture and treat in summer when Bd pressure is lowest, and nearly impossible to capture in winter when antifungal treatment would have the most impact on disease dynamics.

Recommended seasonal forcing: Amplitude **$\sigma_\xi \approx 0.7–0.8$**, with peak in mid-summer (January) and trough in mid-winter (July). Alternatively, a step function or steeper sigmoidal seasonal switch may be more realistic than a smooth cosine, since frog activity has a relatively sharp onset in spring and cessation in autumn.

η (Antifungal Recovery Rate) — Minimally Seasonal

Seasonal range: effectively none

The itraconazole bath protocol is administered in controlled conditions (captive or semi-captive, temperature-controlled water), so the drug's efficacy does not vary seasonally. The 6-day clearance at 0.0025% itraconazole concentration (Brannelly et al. 2012, Dis. Aquat. Org. 101:95–104) is a fixed protocol parameter that produces $\eta \approx 0.167$ day^{-1} regardless of external season.

The only seasonal consideration is that post-treatment re-release in winter means frogs face immediate re-exposure to high Bd pressure (high α, high β), which may reduce the effective population-level benefit of treatment — but this is already captured in the model's structure (treated frogs return to S, where they encounter $\alpha \cdot I \cdot S$ transmission). The parameter η itself should be modeled as **constant year-round**.

Recommended seasonal forcing: none. Keep $\eta = 0.167$ day^{-1} constant.

Seasonal Forcing Priority Ranking

Rank	Parameter	Seasonal range	Amplitude σ	Phase (peak)	Evidence quality
1	**β** (mortality)	**~10×**	0.80–0.90	Mid-winter (Jul)	Direct *L. aurea* experiment (Campbell 2019)
2	**α** (transmission)	~3–5×	0.40–0.60	Late autumn (May–Jun)	Multiple indirect lines; complex phasing

Rank	Parameter	Seasonal range	Amplitude σ	Phase (peak)	Evidence quality
3	**ξ** (capture rate)	~5–8×	0.70–0.80	Mid-summer (Jan)	*L. aurea* field CMR data
4	**ζ** (sauna rate)	~2×	0.30–0.50	Mid-summer (Dec–Jan)	Solar irradiance physics + Waddle 2024
5	**δ** (cadaver decay)	~2–3×	0.30–0.40	Mid-summer (Jan)	Johnson & Speare 2003, 2005
6	**γ** (cadaver transmission)	~2–3×	0.30–0.40	Mid-winter (Jul)	Inverse of δ; no direct *L. aurea* data
7	**η** (antifungal recovery)	None	0	N/A	Protocol is temperature-independent

Implementation Guidance

Minimal seasonal model (add forcing to β only)

If computational simplicity is a priority or you want to test whether seasonality qualitatively changes model behavior before committing to full seasonal complexity, adding forcing to β alone captures the dominant effect:

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β(t) = β₀ · (1 + σ_β · cos(2π(t - t_peak)/365))
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where $\beta_0 \approx 0.012$ day^{-1}, $\sigma_\beta \approx 0.85$, and t_peak = day of peak winter (approximately July 15 in Sydney, or ~day 196 of the calendar year). This gives β oscillating between ~0.002 (January) and ~0.022 (July).

Moderate seasonal model (β + α + ξ)

Adding seasonal forcing to α and ξ alongside β captures the three most impactful seasonal effects — disease severity, transmission pressure, and management feasibility:

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β(t) = β₀ · (1 + 0.85 · cos(2π(t - 196)/365))
α(t) = α₀ · (1 + 0.50 · cos(2π(t - 150)/365))    # peaks ~late May
ξ(t) = ξ₀ · (1 + 0.75 · cos(2π(t - 15)/365))      # peaks ~mid-January
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Note the different phase offsets: β and α peak in winter (with α slightly earlier), while ξ peaks in summer. This creates the key dynamic tension in the model — disease pressure is highest when the capacity to intervene with antifungal baths is lowest.

Full seasonal model (all parameters except η)

For maximum realism, apply seasonal forcing to all six dynamic parameters with the amplitudes and phases in the table above. This adds complexity but may be important for accurately evaluating management strategies that depend on seasonal timing (e.g., deploying saunas before winter vs. conducting antifungal treatment campaigns in late summer before Bd pressure builds).

Key Modeling Insight: The Winter Mortality Bottleneck

The seasonal analysis reveals that *L. aurea* population dynamics under Bd pressure are dominated by a **winter mortality bottleneck**. The confluence of high β, high α, high γ, low δ, low ξ, and reduced ζ effectiveness in winter means that most disease-driven mortality is concentrated in a ~4-month window (May–August). Waddle described this directly: populations struggle to get enough individuals through winter to breed the following spring.

This has profound implications for the optimal control problem:

1. **Frog saunas (ζ) provide their greatest marginal benefit in early autumn** (March–April), before the winter mortality spike, by clearing infections and building the acquired immunity buffer documented in Waddle et al. (2024) before temperatures drop.
2. **Antifungal baths (ξ) face a seasonal access constraint** — they can only be deployed effectively in spring/summer. A **late-summer treatment campaign** (February–March) targeting juveniles before their first winter would maximize impact, consistent with the Cook et al. (2022) finding that treating juvenile *Rana cascadae* before winter produced 4× higher overwinter survival.
3. **The model should reproduce the observed seasonal prevalence pattern** in *L. aurea*: low Bd prevalence in summer, rapid increase in autumn, peak loads and mortality in winter, and recovery in spring among survivors. If the seasonal model does not produce this pattern, the parameter calibration needs adjustment.

Sources

1. Campbell L, Bower DS, Clulow S, et al. (2019) Interaction between temperature and sublethal infection with the amphibian chytrid fungus impacts a susceptible frog species.

Sci. Rep. 9:83. DOI: 10.1038/s41598-018-35874-7

2. Piotrowski JS, Annis SL, Longcore JE (2004) Physiology of *Batrachochytrium dendrobatidis*, a chytrid pathogen of amphibians. *Mycologia* 96:9–15
3. Rachowicz LJ & Briggs CJ (2007) Quantifying the disease transmission function: effects of density on *Batrachochytrium dendrobatidis* transmission in the mountain yellow-legged frog. *J. Anim. Ecol.* 76:711–721
4. Johnson ML & Speare R (2003) Survival of *Batrachochytrium dendrobatidis* in water: quarantine and disease control implications. *Emerg. Infect. Dis.* 9:922–925
5. Johnson ML & Speare R (2005) Possible modes of dissemination of the amphibian chytrid *Batrachochytrium dendrobatidis* in the environment. *Dis. Aquat. Org.* 65:181–186
6. Stockwell MP, Garnham JI, Bower DS, et al. (2016) Low disease-causing threshold in a frog species susceptible to chytridiomycosis. *FEMS Microbiol. Lett.* 363:fnw111
7. Waddle AW, et al. (2024) Hotspot shelters stimulate frog resistance to chytridiomycosis. *Nature* 631:344–349
8. Waddle AW & Shine R. "How to build a frog sauna." Construction guide, The Conversation / Macquarie University
9. Beranek CT, Clulow J, Mahony M (2021) Rapid population increase of the threatened Australian amphibian *Litoria aurea* in response to wetlands constructed as a refuge. *J. Environ. Manage.* 291:112742
10. Brannelly LA, Richards-Zawacki CL, Pessier AP (2012) Clinical trials with itraconazole as a treatment for chytrid fungal infections in amphibians. *Dis. Aquat. Org.* 101:95–104
11. Cook et al. (2022) In situ treatment of juvenile frogs for disease can reverse population declines. *Conserv. Sci. Pract.* 4:e12762
12. Maniero GD & Carey C (1997) Changes in selected aspects of immune function in the leopard frog, *Rana pipiens*, associated with exposure to cold. *J. Comp. Physiol. B* 167:256–263
13. Abu Bakar A, Bower DS, Stockwell MP, et al. (2016) Susceptibility to disease varies with ontogeny and immunocompetence in a threatened amphibian. *Oecologia*. DOI: 10.1007/s00442-016-3607-4